

ECE Probabilistic Systems Analysis

L. Mohammad Riyadh

Department of Communications and
Electronic



Variance of a Random Variable

Let the deviation of the random variable X about its mean be $X - E[X]$

We are interested in the magnitude of the variations only, it is convenient to work with the square of the deviation, which is always positive

$$D(X) = (X - E[X])^2$$

The expected value is a constant, so we will denote it by

$$m_x = E[X]$$

The **variance of the random variable X** is defined as the expected value of D

$$\begin{aligned}\text{VAR}[X] &= E[(X - m_X)^2] \\ &= \sum_{k=1}^{\infty} (x_k - m_X)^2 p_X(x_k) &= E[X^2] - m_X^2\end{aligned}$$

$E[X^2]$ is called the **second moment of X**

Let X be the number of heads in three tosses of a fair coin. Find $\text{VAR}[X]$.

I roll a fair die and let X be the resulting number. Find EX , $\text{Var}(X)$, and σ_X .

We have $R_X = \{1, 2, 3, 4, 5, 6\}$ and $P_X(k) = \frac{1}{6}$ for $k = 1, 2, \dots, 6$. Thus, we have

$$EX = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2};$$

$$EX^2 = 1 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 9 \cdot \frac{1}{6} + 16 \cdot \frac{1}{6} + 25 \cdot \frac{1}{6} + 36 \cdot \frac{1}{6} = \frac{91}{6}.$$

Thus

$$\text{Var}(X) = E[X^2] - (EX)^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} \approx 2.92,$$

$$\sigma_X = \sqrt{\text{Var}(X)} \approx \sqrt{2.92} \approx 1.71$$

Let X be a discrete random variable with the following PMF

$$P_X(k) = \begin{cases} 0.1 & \text{for } k = 0 \\ 0.4 & \text{for } k = 1 \\ 0.3 & \text{for } k = 2 \\ 0.2 & \text{for } k = 3 \\ 0 & \text{otherwise} \end{cases}$$

- a. Find EX . 1.6
- b. Find $\text{Var}(X)$. 0.84
- c. If $Y = (X - 2)^2$, find EY . 1.

Let X be a discrete random variable with $P_X(k) = \frac{1}{5}$ for $k = -1, 0, 1, 2, 3$. Let $Y = 2|X|$.

Find:

1- Range of Y

$$R_Y = \{2|x| \text{ where } x \in R_X\}$$

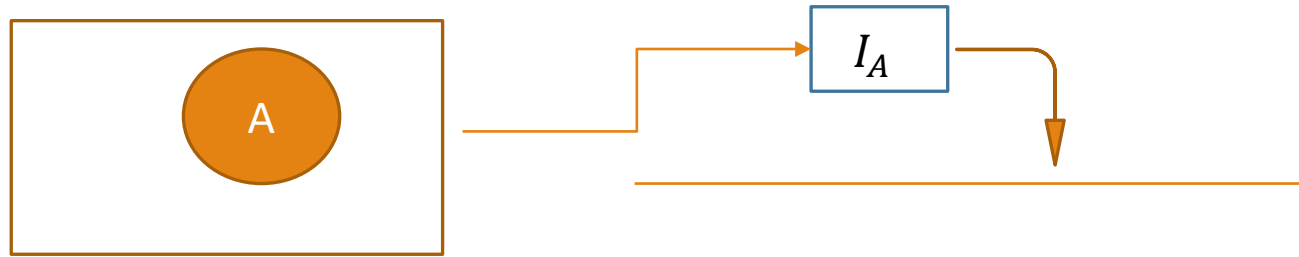
2- PMF of Y

$$= \{0, 2, 4, 6\}.$$

3- $\text{VAR}[Y]$

$$P_Y(k) = \begin{cases} \frac{1}{5} & \text{for } k = 0, 4, 6 \\ \frac{2}{5} & \text{for } k = 2 \\ 0 & \text{otherwise} \end{cases}$$

The Bernoulli Random Variable

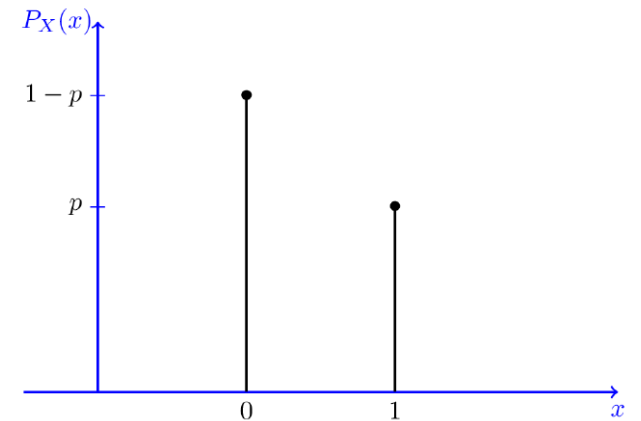


$$I_A = \begin{cases} 1 & \text{if the event } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

A random variable X is said to be a *Bernoulli* random variable with *parameter* p , shown as $X \sim \text{Bernoulli}(p)$, if its PMF is given by

$$P_X(x) = \begin{cases} p & \text{for } x = 1 \\ 1 - p & \text{for } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

where $0 < p < 1$.



The Bernoulli Random Variable

PMF: $p_I(0) = 1 - p$ and $p_I(1) = p$,

Mean: $m_I = E[I_A] = p$

Variance: $\text{VAR}[I_A] = p(1 - p) = pq$

Binomial Random Variable

A random variable X is said to be a *binomial* random variable with parameters n and p , shown as $X \sim \text{Binomial}(n, p)$, if its PMF is given by

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

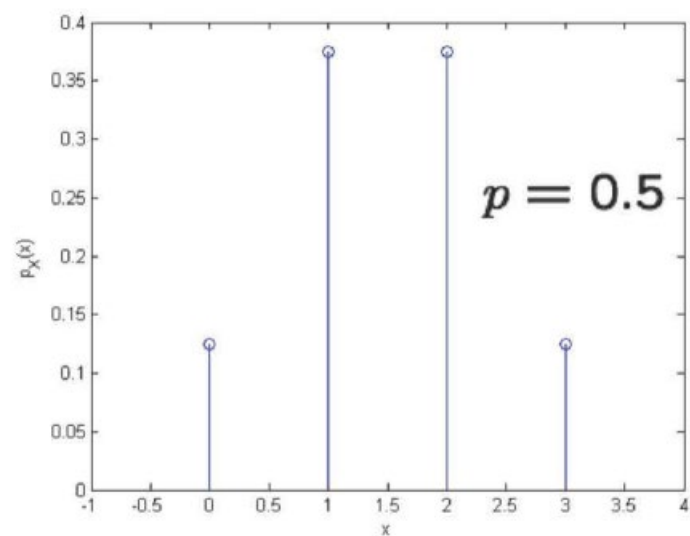
where $0 < p < 1$.

Experiment: n independent tosses of a coin with $\mathbf{P}(\text{Heads}) = p$

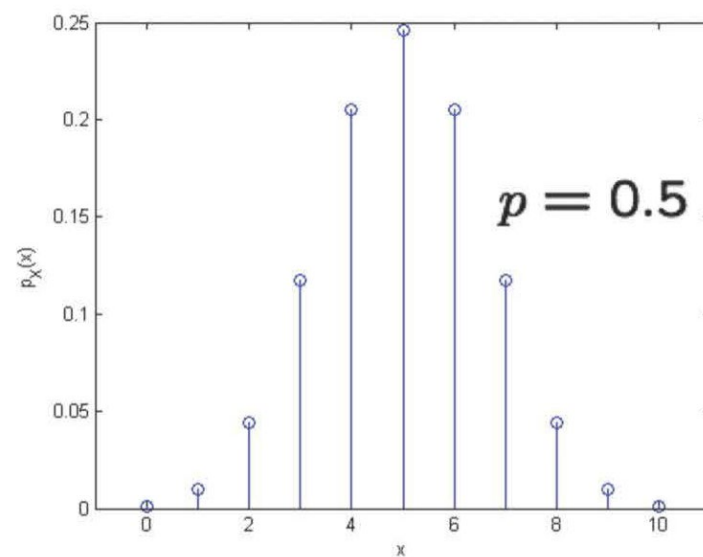
Sample space: Set of sequences of H and T, of length n

Random variable X : number of Heads observed

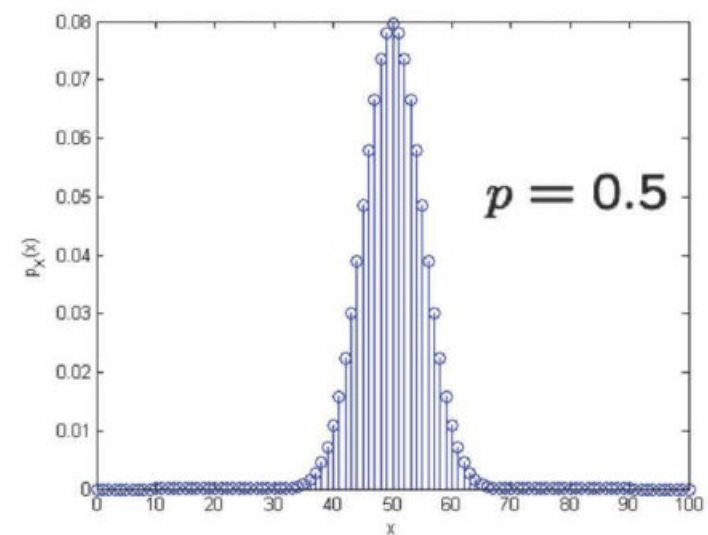
$n = 3$



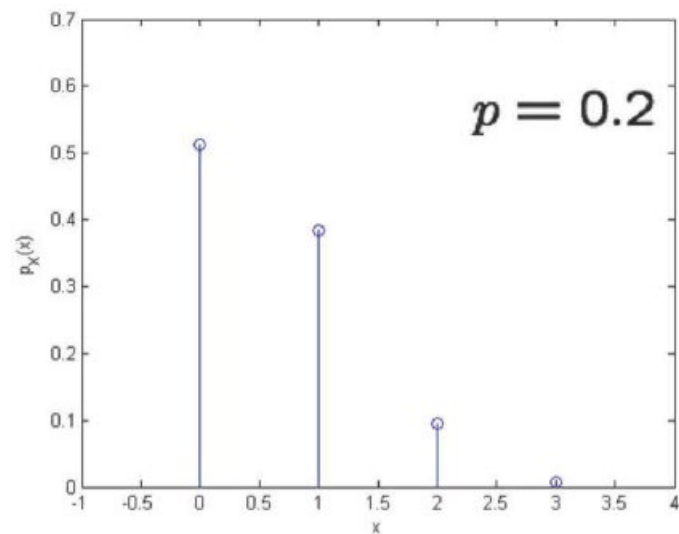
$n = 10$



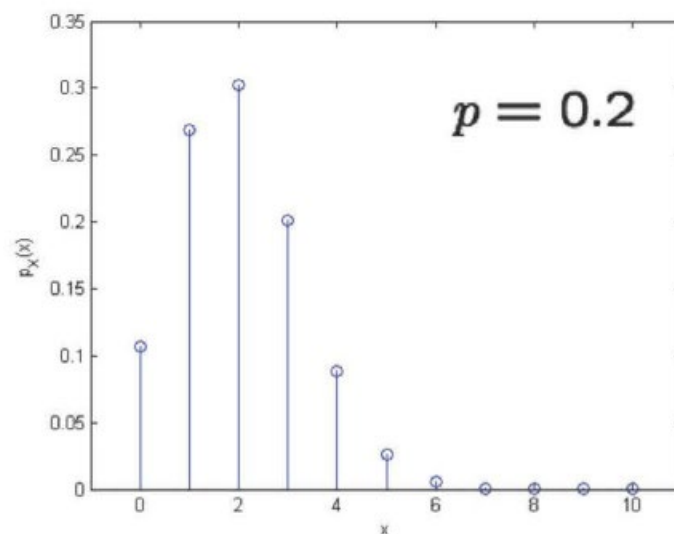
$n = 100$



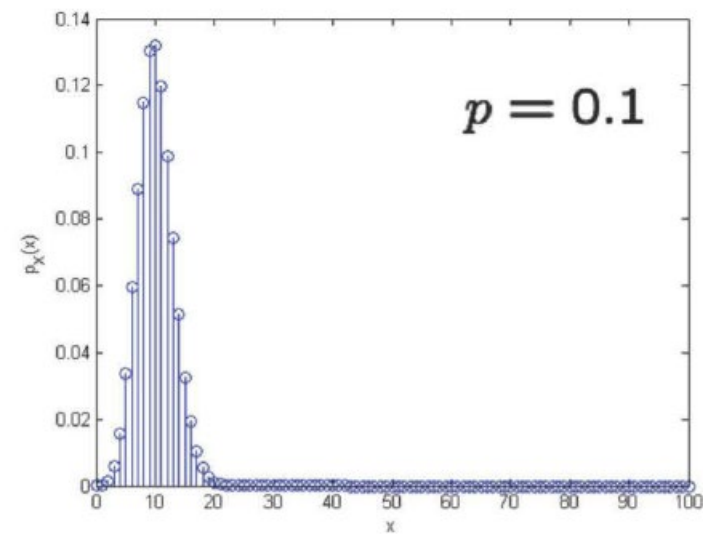
$p = 0.2$



$p = 0.2$



$p = 0.1$



Binomial Random Variable

Mean of a Binomial Random Variable

$$\begin{aligned} E[X] &= \sum_{k=0}^n k p_X(k) = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=1}^n k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\ &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k} \\ &= np \sum_{j=0}^{n-1} \frac{(n-1)!}{j!(n-1-j)!} p^j (1-p)^{n-1-j} = np \end{aligned}$$

$$\boxed{E[X] = np}$$

Binomial Random Variable

Variance of a Binomial Random Variable

$$\begin{aligned} E[X^2] &= \sum_{k=0}^n k^2 \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} = \sum_{k=1}^n k \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k} \\ &= np \sum_{k'=0}^{n-1} (k'+1) \binom{n-1}{k'} p^{k'} (1-p)^{n-1-k'} \\ &= np \left\{ \sum_{k'=0}^{n-1} k' \binom{n-1}{k'} p^{k'} (1-p)^{n-1-k'} + \sum_{k'=0}^{n-1} 1 \binom{n-1}{k'} p^{k'} (1-p)^{n-1-k'} \right\} \\ &= np \{ (n-1)p + 1 \} = np(np + q). \end{aligned}$$

$$\sigma_X^2 = E[X^2] - E[X]^2 = np(np + q) - (np)^2 = npq = np(1 - p).$$

Geometric Random Variable

I have a coin with $P(H) = p$

I toss the coin until I observe the first heads

A random variable X is said to be a *geometric* random variable with *parameter* p , shown as $X \sim \text{Geometric}(p)$, if its PMF is given by

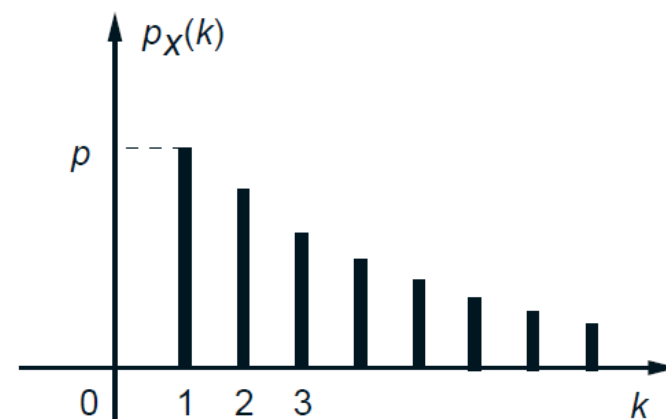
$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

where $0 < p < 1$.

$$R_X = \{1, 2, 3, \dots\}$$

The sum of the geometric probabilities is 1.

$$\sum_{k=1}^{\infty} p_X(k) = \sum_{k=1}^{\infty} (1-p)^{k-1} p = p \sum_{k=0}^{\infty} (1-p)^k = p \cdot \frac{1}{1-(1-p)} = 1.$$



Geometric Random Variable

Mean of a Geometric Random Variable

$$E[X] = \sum_{k=1}^{\infty} k p q^{k-1} = p \sum_{k=1}^{\infty} k q^{k-1}.$$

$$E[X] = p \frac{1}{(1 - q)^2} = \frac{1}{p}.$$

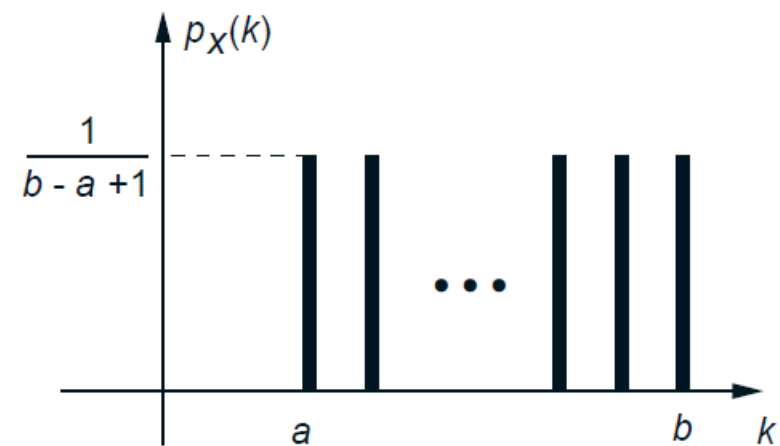
$$\frac{1}{1 - x} = \sum_{k=0}^{\infty} x^k$$

$$\frac{1}{(1 - x)^2} = \sum_{k=0}^{\infty} k x^{k-1}.$$

Variance of the geometric random variable

$$\text{VAR}[X] = \frac{1 - p}{p^2}$$

Discrete Uniform Random Variable



$$p_X(k) = \begin{cases} \frac{1}{b-a+1} & \text{if } k = a, a+1, \dots, b, \\ 0 & \text{otherwise,} \end{cases}$$

$$\mathbf{E}[X] = \frac{a+b}{2},$$

$$\text{var}(X) = \frac{(b-a)(b-a+1)}{12}.$$

Transmission Errors

A binary communications channel introduces a bit error in a transmission with probability p . Let X be the number of errors in n independent transmissions. Find the pmf of X . Find the probability of one or fewer errors.

If $n = 1000$ bit per transmit and $p = 0.001$, what is the probability of three errors or less

Each transmission results in a “0” if there is no error and a “1” if there is an error,

$$P[1] = P$$

$$P[0] = 1 - P$$

The probability of k errors in n bit transmissions is given by

$$p_X(k) = P[X = k] = \binom{n}{k} p^k (1 - p)^{n-k} \quad k = 0, 1, \dots, n.$$

We call X the **binomial random variable**, with parameters n and p .

$$P[X \leq 1] = \binom{n}{0} p^0 (1 - p)^{n-0} + \binom{n}{1} p^1 (1 - p)^{n-1} = (1 - p)^n + np(1 - p)^{n-1}.$$

Message Transmissions

Let X be the number of times a message needs to be transmitted until it arrives correctly at its destination. Find the pmf of X . Find the probability that X is an even number.

X is a discrete random variable taking on values from $R_X = \{1, 2, 3, 4, 5, \dots\}$

The event $\{X = k\}$ occurs if the underlying experiment finds $k - 1$ consecutive erroneous transmissions (“failures”) followed by a error-free one (“success”):

$$p_X(k) = P[X = k] = P[00 \dots 01] = (1 - p)^{k-1}p = q^{k-1}p \quad k = 1, 2, \dots$$

We call X the **geometric random variable**, and we say that X is geometrically distributed.

$$P[X \text{ is even}] = \sum_{k=1}^{\infty} p_X(2k) = p \sum_{k=1}^{\infty} q^{2k-1} = p \frac{1}{1 - q^2} = \frac{1}{1 + q}.$$

An urn contains 9 \$1 bills and one \$50 bill. Let the random variable X be the total amount that results when two bills are drawn from the urn without replacement.

- A. Describe the underlying space S of this random experiment and specify the probabilities of its elementary events.
 - B. Show the mapping from S to R_X the range of X .
 - C. Find the probabilities for the various values of X .
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A modem transmits a +2voltage signal into a channel. The channel adds to this signal a noise term that is drawn from the set $\{0, -1, -2, -3\}$ with respective probabilities $\{4/10, 3/10, 2/10, 1/10\}$.

- A. Find the pmf of the output Y of the channel.
 - B. What is the probability that the output of the channel is equal to the input of the channel?
 - C. What is the probability that the output of the channel is positive?
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A modem transmits over an error-prone channel, so it repeats every “0” or “1” bit transmission five times. We call each such group of five bits a “codeword.” The channel changes an input bit to its complement with probability $p = 1/10$ and it does so independently of its treatment of other input bits. The modem receiver takes a majority vote of the five received bits to estimate the input signal. Find the probability that the receiver makes the wrong decision.

- 3.46.** Heat must be removed from a system according to how fast it is generated. Suppose the system has eight components each of which is active with probability 0.25, independently of the others. The design of the heat removal system requires finding the probabilities of the following events:
- (a)** None of the systems is active.
 - (b)** Exactly one is active.
 - (c)** More than four are active.
 - (d)** More than two and fewer than six are active.